



Daniel Mouzo Calzadilla

Senior Bioinformatics & Workflow Architect

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Senior Bioinformatician with 10+ years of combined experience translating complex genomic datasets into scalable, reproducible pipelines. Expert in Nextflow workflow engineering, on-prem HPC cluster management, and leading cross-functional collaborations for €4M international consortia. Fully prepared to design cloud-ready analytical workflows under strict quality and data integrity standards.

WORK EXPERIENCE

Senior Bioinformatician, BCI, Queen Mary University of London

2025 - Present

- Engineered production-grade Nextflow pipelines processing 20+ TB of clinical WGS data within the Genomics England.
- Automated reporting systems, integrating variant calling with SQL clinical data to reduce turnaround times to under 3 hours.

☆ Workflow Engineering, Big Data Architecture, Nextflow, Genomic England, Shiny dashboards

Visiting Scientist, Gene Therapy, Ospedale San Raffaele

2025

- Harmonized scMultiome datasets across XPAND clinical sites to support cross-centre comparison and downstream translational analysis.
- Validated LT-HSC signatures in AML-derived cells, linking multi-omic profiles to clinically relevant cell-state interpretation.

☆ Data Standardization, Single-Cell, Gene therapy, Data Provenance

Visiting Scientist, Workflow & Data Systems, KAUST

2023

- Designed Snakemake preprocessing workflows for large-scale NGS datasets, reducing turnaround time and standardizing execution.
- Built relational databases to centralize experimental metadata and strengthen data integrity across collaborative projects.

☆ Relational Database Design, Snakemake, Quality Control

Postdoctoral Bioinformatician, Navarrabiomed

2021- 2025

- Coordinated a cross-functional work package of 30+ multi-site researchers for European gene therapy project
- Developed automated single-cell NGS pipelines, standardizing and data harmonization across international clinical centers.

☆ Gene Regulatory Networks, Data Harmonization, NGS pipelines, Consortium Management

Research Assistant, University Santiago de Compostela

2015-2020

- Optimized LC-MS and quantitative proteomics workflows, reducing turnaround time by 1.5 days and increasing sample throughput.
- Developed traceability analytics for industrial quality control, translating experimental data into cost-saving decisions.

☆ Mass Spectrometry, Industrial Biotechnology, Two-dimensional electrophoresis

CERTIFICATES

ISO 13485 Internal Auditor

Comply Guru

2026

AWS Cloud Practitioner

AWS

2025

Nextflow Advanced

Sequera

2025

Project Management

Google

2023

Machine learning and Big Data for Bioinformatics

University of Granada

2023

Cybersecurity

Cámara de Comercio

2023

Data Science Specialized Program

IBM

2020

EDUCATION

PhD in Molecular Biology

University of Santiago de Compostela

2017-2020

Master Degree in Innov. in Food Safety and Tech.

University of Santiago de Compostela

2016-2020

Degree in Biology

University of Santiago de Compostela

2012-2016

SELECTED PROJECTS

scAnnex and bulkAnnex

2026

- scAnnex: Built an automated Nextflow pipeline and R Shiny dashboard to standardize preprocessing, adaptive QC, and cell-type annotation for scRNA-seq datasets.
- bulkAnnex: Reproducible Nextflow workflow for bulk RNA-seq that integrates automated quality control, differential expression, and multi-database pathway enrichment.

Genomics England Variant Engine CRUK Project

2025

- Built an automated Nextflow pipeline for clinical variant calling, and high-throughput reporting.

XPAND EU Project

2024

- Harmonized scMultiome clinical datasets across 3 European hospitals for therapeutic HSC characterization. Patent pending.

LEADERSHIP & TEACHING

Designed and taught courses for 100+ researchers

- scMultiome analysis and integration: workflow management (PALS Summer School) [2026]
- Data Analysis with R: Introduction and Basic Statistics (Navarrabiomed) [2024]
- Introduction to Omics Data Analysis: Transcriptomics (Navarrabiomed) [2024]
- Founded and coordinated 2 cross-disciplinary biotechnology book clubs. (Kaust) [2023]

HARD SKILLS

- R
- Linux
- Python
- SQL
- Nextflow
- Git
- Machine Learning
- Big Data
- Containerization
- AWS
- Project Management

SOFT SKILLS

- Adaptability
- Cross-Functional Collaboration
- Problem-solving
- Agile Management
- Scientific Communication
- Leadership

LANGUAGES

- Spanish
- English
- Galician